SPY / IEF Trend Rotation

A rules-based approach to managing equity drawdowns

Prepared October 19, 2026

Hypothetical backtested results. For discussion purposes only. Generated by an AI multi-agent research pipeline.

Executive Summary

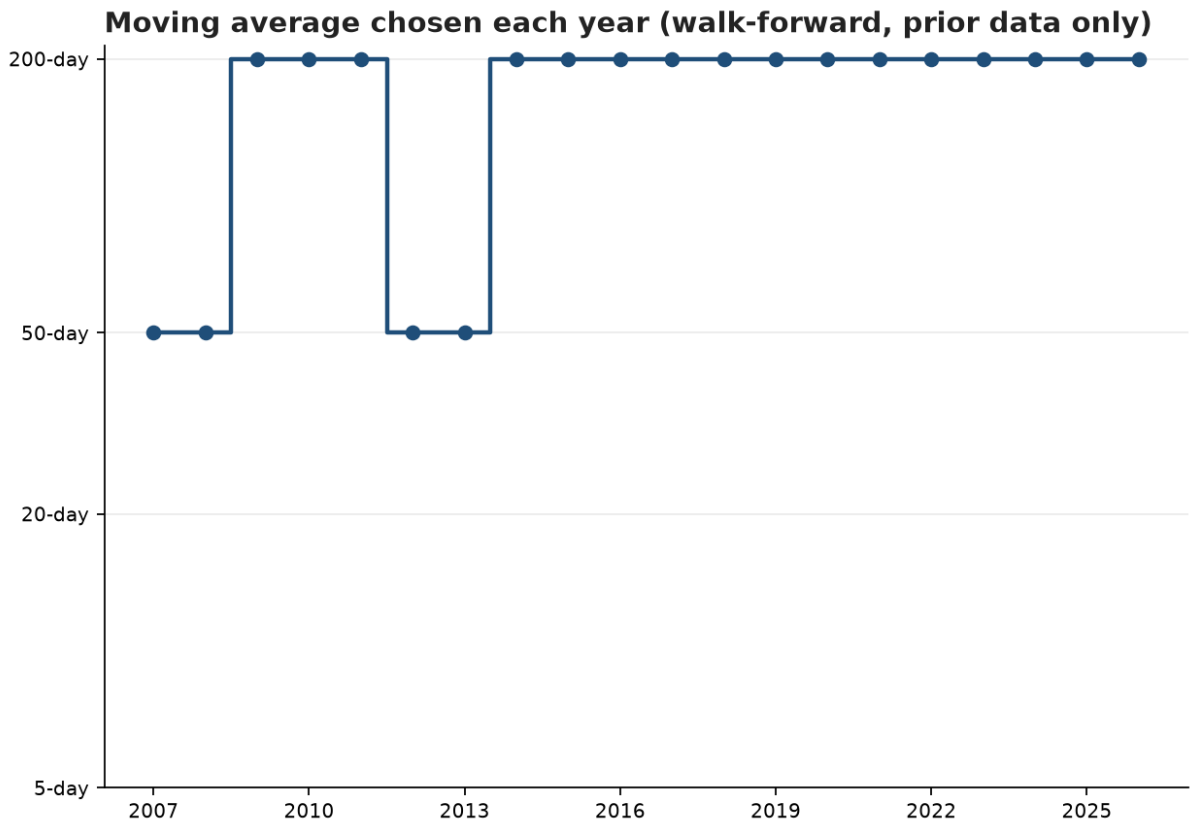
- A tactical SPY/Treasury allocation strategy: holds SPY above a moving average, IEF otherwise, with the average re-selected each January using only prior data
- Since 2007, max drawdown was roughly half of SPY's (-26.4% vs -55.2%), with lower volatility (12.4% vs 19.5%)
- The trade-off is return: 7.6% CAGR vs 10.9% for SPY since 2007, and a Sharpe ratio of 0.53 vs 0.55
- Drawdown protection has not held in every period: over 5Y the strategy's max drawdown was deeper than SPY's (-26.4% vs -24.5%)
- Best suited as a risk-managed equity sleeve for investors who value shallower deep drawdowns over full upside capture

The Problem: Deep Equity Drawdowns

- Since 2007, SPY's maximum drawdown was -55.2%, and its worst rolling 1-year return was -47.4%
- In 2008, SPY returned -36.8%; deep losses like this can force selling or derail long-term plans
- SPY's volatility since 2007 was 19.5%, and its worst single day was -10.9%
- Investors seeking equity exposure often want a rules-based way to step aside during sustained downtrends

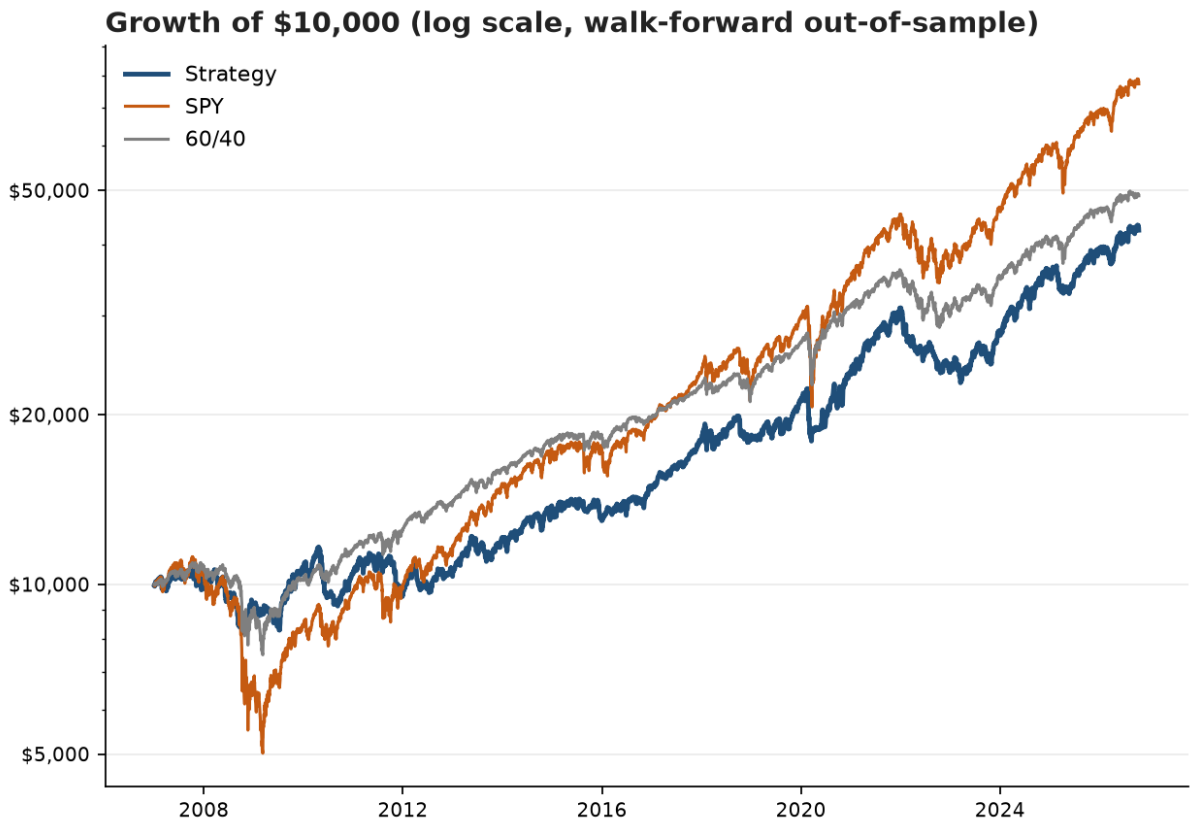
How the Strategy Works

- Daily rule: hold SPY when SPY closes above its moving average; otherwise hold IEF (intermediate Treasuries)
- Candidate moving averages: 5, 20, 50 or 200 days
- Each January the average is re-selected using only data available at that time (walk-forward), avoiding hindsight in the results
- Over 20 years of walk-forward selection, the 200-day was chosen 16 times and the 50-day 4 times
- Since 2007 the strategy held SPY 77.0% of the time, with 158 switches; it is currently 100.0% in SPY



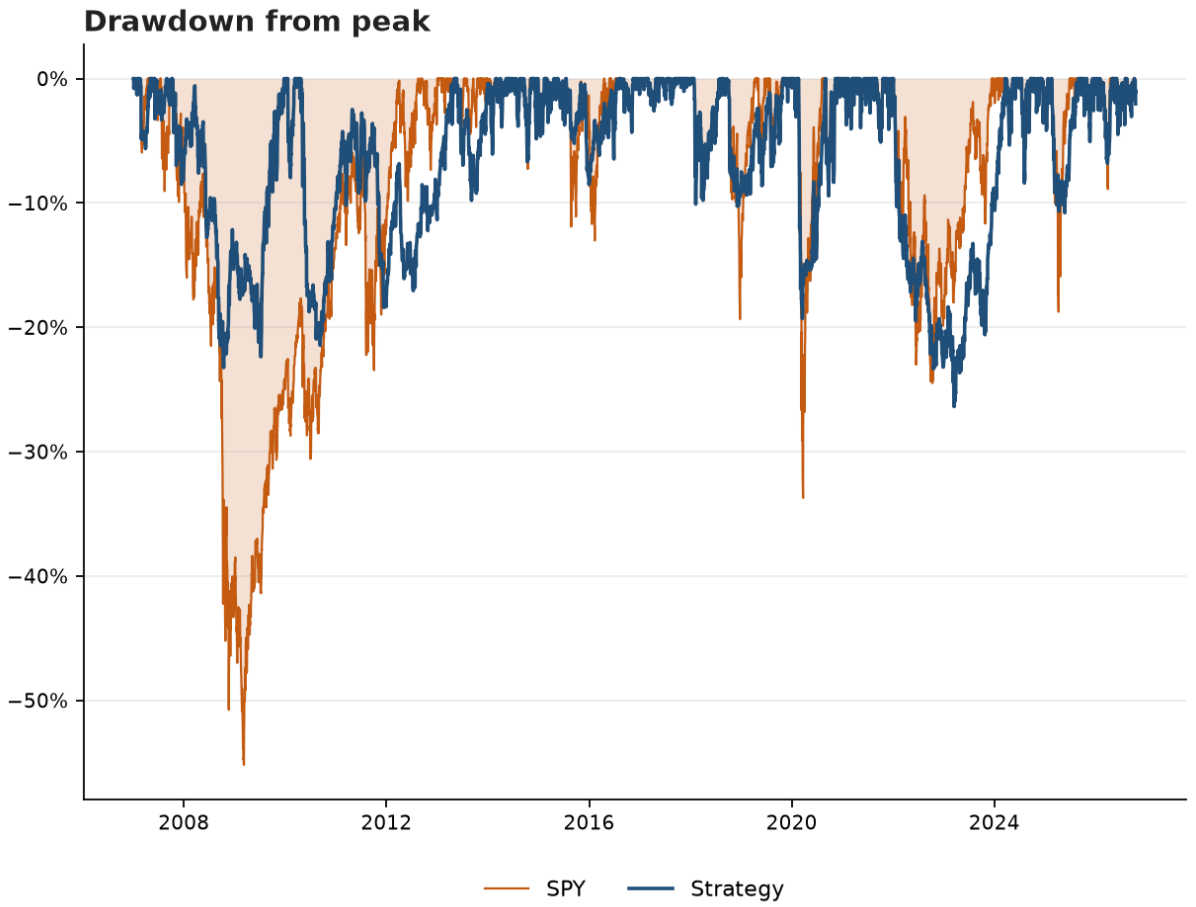
Performance vs SPY Since 2007

- $10,000 invested in January 2007 grew to $42,495, vs $77,253 for SPY and $48,905 for a 60/40 portfolio
- Total return of 324.9% vs 672.5% for SPY; CAGR of 7.6% vs 10.9%
- Lower volatility than SPY (12.4% vs 19.5%) and beta of 0.29
- Up capture of 54.9% vs down capture of 53.5%: the strategy captures only part of SPY's upside as well as its downside



Risk and Drawdown Protection

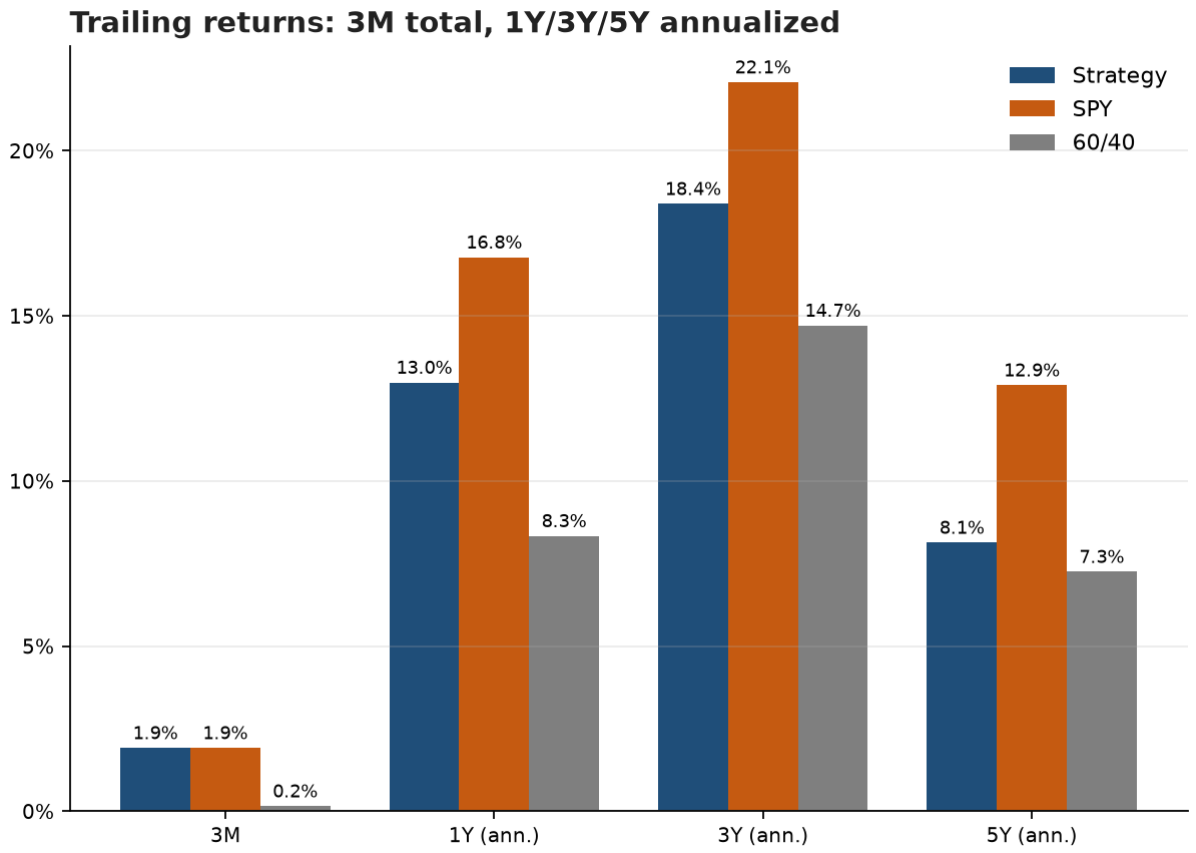
- Since 2007, lower max drawdown than SPY (-26.4% vs -55.2%)
- Worst rolling 1-year return of -23.1% vs -47.4% for SPY; best rolling 1-year of 44.6% vs 77.5% for SPY
- In 2008 the strategy returned -6.8% vs -36.8% for SPY; in 2018, 1.6% vs -4.6%
- Calmar ratio since 2007 of 0.29 vs 0.20 for SPY
- Protection is not uniform: in 2022 the strategy returned -22.7% vs -18.2% for SPY, as stocks and Treasuries fell together



Hypothetical backtested performance, net of modeled trading costs, not of fees. Not actual trading. Past performance does not guarantee future results. See Important Disclosures.

Performance by Period: 3M, 1Y, 3Y, 5Y

- 3M: fully invested in SPY, so results matched SPY: 1.9% return and -3.1% max drawdown for both
- 1Y: 13.0% return vs 16.8% for SPY; max drawdown -6.8% vs -8.9%; Sharpe 0.78 vs 1.00
- 3Y: 18.4% CAGR vs 22.1% for SPY; max drawdown -10.8% vs -18.8%; Sharpe 1.11 vs 1.10
- 5Y: 8.1% CAGR vs 12.9% for SPY; max drawdown -26.4% vs -24.5%; Sharpe 0.39 vs 0.57
- The strategy trailed SPY on return in every window above except 3M, where it matched



Risk and Return by Period: Strategy vs SPY

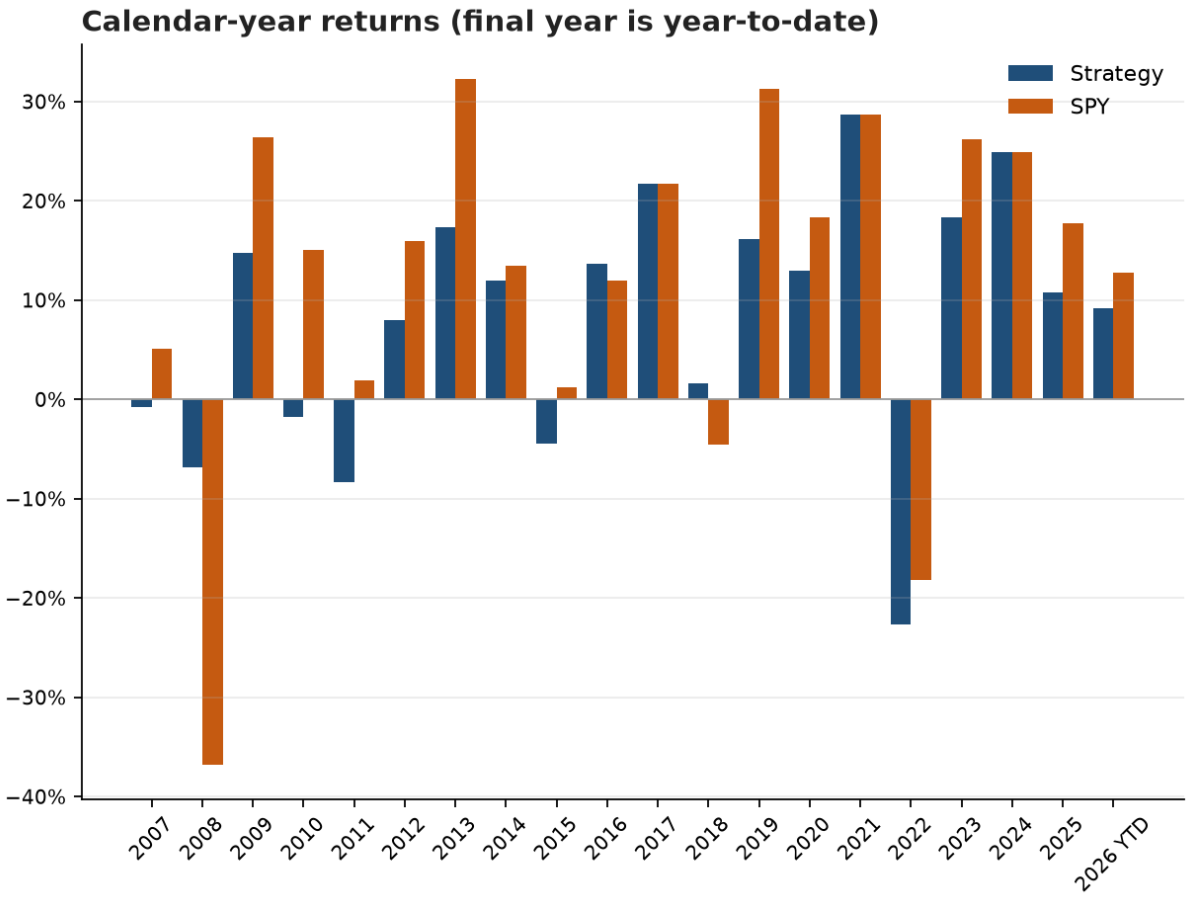
Metric	3M Strat.	3M SPY	1Y Strat.	1Y SPY	3Y Strat.	3Y SPY	5Y Strat.	5Y SPY	SI Strat.	SI SPY
Return (3M total, else annualized)	**1.9%**	**1.9%**	**13.0%**	**16.8%**	**18.4%**	**22.1%**	**8.1%**	**12.9%**	**7.6%**	**10.9%**
Total return (cumulative)	1.9%	1.9%	13.0%	16.8%	65.6%	81.5%	47.6%	83.0%	324.9%	672.5%
Volatility (ann.)	11.1%	11.1%	11.9%	12.7%	12.0%	15.3%	12.3%	17.1%	12.4%	19.5%
Sharpe ratio	n/a	n/a	0.78	1.00	1.11	1.10	0.39	0.57	0.53	0.55
Sortino ratio	n/a	n/a	1.11	1.47	1.57	1.63	0.54	0.83	0.72	0.77
Max drawdown	-3.1%	-3.1%	-6.8%	-8.9%	-10.8%	-18.8%	-26.4%	-24.5%	-26.4%	-55.2%
Calmar ratio	n/a	n/a	1.90	1.89	1.70	1.18	0.31	0.53	0.29	0.20
Beta vs SPY	1.00	1.00	0.87	1.00	0.58	1.00	0.46	1.00	0.29	1.00
Correlation vs SPY	1.00	1.00	0.94	1.00	0.74	1.00	0.65	1.00	0.45	1.00
Up capture	100.0%	100.0%	91.7%	100.0%	83.3%	100.0%	65.7%	100.0%	54.9%	100.0%
Down capture	100.0%	100.0%	94.6%	100.0%	83.3%	100.0%	65.9%	100.0%	53.5%	100.0%
% days positive	44.6%	44.6%	52.6%	52.6%	56.2%	56.4%	53.6%	53.7%	54.1%	55.0%
Best day	1.8%	1.8%	1.9%	2.9%	2.5%	10.5%	2.5%	10.5%	4.4%	14.5%
Worst day	-1.5%	-1.5%	-2.6%	-2.6%	-3.0%	-5.9%	-3.0%	-5.9%	-5.8%	-10.9%
Switches / % time in SPY	0 / 100.0%		2 / 95.2%		8 / 92.2%		30 / 78.2%		158 / 77.0%	

60/40 benchmark	3M	1Y	3Y	5Y	SI
Return (3M total, else ann.)	0.2%	8.3%	14.7%	7.3%	8.4%
Total return (cumulative)	0.2%	8.3%	50.7%	41.7%	389.0%
Volatility (ann.)	7.8%	8.5%	9.7%	11.0%	11.2%
Sharpe ratio	n/a	0.55	1.02	0.35	0.63
Max drawdown	-2.8%	-6.0%	-10.6%	-21.0%	-31.4%
Calmar ratio	n/a	1.39	1.39	0.35	0.27

Periods end on the last completed session. 3M: 2026-07-17 to 2026-10-16, 1Y: 2025-10-17 to 2026-10-16, 3Y: 2023-10-17 to 2026-10-16, 5Y: 2021-10-18 to 2026-10-16, SI: 2007-01-03 to 2026-10-16. SI = since the walk-forward start. Ratios use the 13-week T-bill as the risk-free rate. Current allocation: 100.0% SPY.

When It Lags: Honest Trade-offs

- 1Y: worse than SPY on Sharpe ratio (0.78 vs 1.00)
- 5Y: worse than SPY on max drawdown (-26.4% vs -24.5%) and on Sharpe ratio (0.39 vs 0.57), driven largely by 2022 (-22.7% vs -18.2%)
- Since inception (2007): worse than SPY on Sharpe ratio (0.53 vs 0.55), with lower CAGR (7.6% vs 10.9%)
- Beat SPY in only 18.7% of rolling 1-year periods; median rolling 1-year excess return was -4.1%
- Whipsaws in choppy markets hurt: e.g. 2011 (-8.3% vs 1.9% for SPY) and 2015 (-4.5% vs 1.2%)
- Strong rallies are only partly captured: 2019 returned 16.2% vs 31.2% for SPY



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Portfolio Fit

- Suits investors who want equity participation with a rules-based mechanism to reduce exposure in sustained downtrends
- Can complement a core SPY or 60/40 holding: correlation to SPY of 0.45 and beta of 0.29 since 2007
- Since 2007, a 60/40 portfolio returned 8.4% CAGR with -31.4% max drawdown, vs 7.6% and -26.4% for the strategy
- Less suited to investors focused on maximizing long-run return or tracking SPY closely
- The Treasury leg can lose money when rates rise, as in 2022, so drawdown protection is not assured

Context: Each Moving Average Over the Full Period (IN-SAMPLE)

Rule (IN-SAMPLE)	CAGR	Volatility	Sharpe	Max DD	Switches	% in SPY	Years chosen
5-day MA switch	-3.5%	12.9%	-0.33	-66.5%	1330	59.4%	0
20-day MA switch	4.0%	12.3%	0.26	-37.1%	570	66.3%	0
50-day MA switch	6.0%	12.0%	0.42	-29.8%	348	70.2%	4
200-day MA switch	9.4%	12.5%	0.66	-26.4%	112	79.1%	16

IN-SAMPLE (2007-01-03 to 2026-10-16): these rows use hindsight. Picking the best row after the fact would overstate results. The strategy instead picks a moving average each January using only earlier data, and that walk-forward result is what the performance pages show. "Years chosen" counts how many years the walk-forward selected each rule.

Next Steps

- Request the full methodology, including walk-forward selection rules and data sources
- Review the supporting-data appendix and calendar-year results
- Discuss sizing and how the strategy might sit alongside existing equity and bond holdings
- Arrange a due-diligence session with the investment team

For discussion purposes only. Not an offer or solicitation.

Meet the Digital Investment Team

Margaret Okafor-Lindqvist

Lead Portfolio Manager

Focus: Overall strategy oversight, allocation policy and investment committee leadership

Certifications: CFA, CAIA | **Experience:** 28 yrs

Education

MBA, Finance, University of Chicago Booth School of Business (2003)
BA, Economics, Northwestern University (1997)

Experience

Senior Portfolio Manager, Tactical Allocation, Harborview Capital Partners (2012-2025)
Portfolio Manager, Ridgeline Asset Advisors (2005-2012)
Investment Analyst, Lakeshore Fiduciary Group (1998-2005)

Margaret Okafor-Lindqvist is an AI-generated fictional persona who leads the team as Portfolio Manager. Her simulated career spans nearly three decades in multi-asset and tactical allocation roles at invented firms. On this team she owns the investment process end to end, chairing reviews of signal changes, approving risk limits and ensuring the rotation between equities and Treasuries stays faithful to its documented rules. Her distinctive strength is disciplined governance: she insists every rule change be justified in writing before implementation, keeping the strategy systematic rather than discretionary.

AI-generated fictional persona

Daniel Ferreira-Holt

Senior Quantitative Research Analyst

Focus: Quantitative research and moving-average signal design

Certifications: CQF | **Experience:** 15 yrs

Education

PhD, Applied Mathematics, Cornell University (2011)
BS, Physics, University of Texas at Austin (2005)

Experience

Quantitative Researcher, Quarrystone Quantitative (2015-2025)
Research Associate, Alderbank Systematic Strategies (2011-2015)

Daniel Ferreira-Holt is an AI-generated fictional persona serving as the team's quantitative research lead. Trained as an applied mathematician, his invented background includes years building trend-following models at fictional systematic shops. He owns the design and testing of the moving-average crossover rules, including lookback selection, whipsaw filters and robustness checks across parameter neighborhoods. His distinctive strength is skepticism toward overfitting; he routinely stress-tests signals on out-of-sample periods and alternative parameter sets before recommending any refinement to the Portfolio Manager.

AI-generated fictional persona

Priya Natarajan

Risk Management Analyst

Focus: Portfolio risk, drawdown monitoring and stress testing

Certifications: FRM, PRM | **Experience:** 12 yrs

Education

MS, Financial Engineering, Columbia University (2014)
BTech, Electrical Engineering, Indian Institute of Technology Madras (2012)

Experience

Risk Analyst, Silvergate Point Advisors (2018-2025)
Risk Associate, Meridian Crest Investments (2014-2018)

Priya Natarajan is an AI-generated fictional persona responsible for risk management on the team. With an engineering background and simulated experience in risk functions at invented asset managers, she owns volatility monitoring, drawdown analysis and scenario testing for both the equity and Treasury legs. She also evaluates periods when stocks and bonds fall together, a key vulnerability for rotation strategies. Her distinctive strength is translating complex risk metrics into clear, actionable dashboards that the Portfolio Manager and investors can interpret quickly.

AI-generated fictional persona

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Meet the Digital Investment Team (continued)

Thomas Reinholt

Macro and Rates Analyst

Focus: Interest rates, Treasury duration and the IEF allocation

Certifications: CFA | **Experience:** 17 yrs

Education

MA, Economics, Georgetown University (2009)
BA, History, University of Wisconsin-Madison (2006)

Experience

Rates Strategist, Bellmont Fixed Income Partners (2016-2025)
Economic Research Analyst, Cobalt Harbor Research (2009-2016)

Thomas Reinholt is an AI-generated fictional persona covering macroeconomics and interest rates. After studying history and economics, his invented career took him through fixed-income research and rates strategy at fictional firms. He owns analysis of the intermediate Treasury side of the rotation, tracking yield-curve dynamics, inflation regimes and how duration exposure behaves when the strategy moves defensive. His distinctive strength is historical perspective: he contextualizes current rate environments against past cycles, helping the team explain when the bond leg may offer weaker protection.

AI-generated fictional persona

Kenji Alvarez-Mori

Market Data Engineer

Focus: Market data engineering and backtest integrity

Certifications: None | **Experience:** 8 yrs

Education

BS, Computer Science, University of Washington (2018)

Experience

Data Engineer, Tidewater Data Systems (2021-2025)
Junior Software Engineer, Pinecrest Analytics (2018-2021)

Kenji Alvarez-Mori is an AI-generated fictional persona who serves as the team's market data engineer. A computer scientist by training, his invented background includes building data pipelines at fictional analytics firms. He owns price and dividend data ingestion, total-return adjustments and the backtesting codebase, guarding against look-ahead bias, survivorship issues and misaligned timestamps. His distinctive strength is reproducibility: every backtest he produces can be rerun from versioned data and code, giving the team confidence that results reflect the rules rather than data errors.

AI-generated fictional persona

Claire Beaumont

Trading and Investor Communications Analyst

Focus: Trade execution, rebalancing operations and investor communications

Certifications: CMT | **Experience:** 6 yrs

Education

BBA, Finance, University of Michigan (2020)
BA, Communications, University of Michigan (2020)

Experience

Execution Associate, Northfield Brokerage Services (2022-2025)
Client Service Analyst, Ashgrove Wealth Partners (2020-2022)

Claire Beaumont is an AI-generated fictional persona handling trading and investor communications. Her invented career began in client service before moving to an execution desk at a fictional brokerage. She owns implementation of signal-driven rotations, minimizing slippage around rebalance dates, and drafts plain-language updates explaining each switch between equities and Treasuries. Her distinctive strength is bridging operations and clients: she can explain exactly what the strategy did and why, without jargon, while keeping execution tightly aligned with the rules.

AI-generated fictional persona

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Year	Strategy	SPY	60/40
2007	-0.7%	5.1%	7.7%
2008	-6.8%	-36.8%	-16.7%
2009	14.7%	26.4%	13.3%
2010	-1.8%	15.1%	13.4%
2011	-8.3%	1.9%	8.3%
2012	8.0%	16.0%	11.3%
2013	17.4%	32.3%	15.6%
2014	12.0%	13.5%	12.0%
2015	-4.5%	1.2%	1.8%
2016	13.6%	12.0%	7.8%
2017	21.7%	21.7%	13.8%
2018	1.6%	-4.6%	-2.0%
2019	16.2%	31.2%	21.8%
2020	12.9%	18.3%	16.9%
2021	28.7%	28.7%	15.1%
2022	-22.7%	-18.2%	-16.4%
2023	18.3%	26.2%	17.0%
2024	24.9%	24.9%	14.2%
2025	10.8%	17.7%	14.3%
2026 YTD	9.1%	12.8%	6.3%

Rolling 1-year returns	Strategy	SPY
Best 1-year period	44.6%	77.5%
Worst 1-year period	-23.1%	-47.4%
Periods beating SPY	18.7%	
Median excess vs SPY	-4.1%	
Growth of $10,000 (since walk-forward start)		
Strategy	$42,495	
SPY	$77,253	
60/40	$48,905	

Calendar-year returns are total returns. The final year is year-to-date through 2026-10-16. Rolling figures cover every 252-trading-day window since the walk-forward start. 60/40 is rebalanced daily, without costs.

Important Disclosures

Hypothetical performance. All performance shown is hypothetical and backtested. It does not represent actual trading or actual client accounts. Backtested results are prepared with the benefit of hindsight and have inherent limitations. No representation is made that any account will or is likely to achieve similar results.

Methodology. The strategy holds either SPY or IEF, based on whether SPY closed above a moving average the previous day. The moving average (5, 20, 50 or 200 days) is chosen each January using only data available before that date (walk-forward). Results include a modeled trading cost of 5 basis points per full switch between SPY and IEF. They exclude management fees, taxes and market impact beyond that assumption, and use adjusted closing prices, which include dividends. Partial calendar years are year-to-date.

In-sample results. Comparisons of individual moving-average rules over the full period are in-sample. They are shown for context only, and overstate what an investor could have known in advance.

Risk. Investing involves risk, including loss of principal. Trend-following signals can lag reversals, cause frequent switching in choppy markets, and underperform buy-and-hold for long periods. Treasury ETFs carry interest-rate risk and can fall alongside equities.

Data. Prices from Yahoo Finance via the yfinance library. The data has not been independently verified. The risk-free rate is the 13-week Treasury bill yield (^IRX).

Investment team. The investment team members in this presentation are AI-generated fictional personas created by an AI system for illustration. They are not real people, and their stated employers are fictional.

Not advice. This presentation was produced by an automated multi-agent research prototype. It is for informational purposes only, and is not investment advice, an offer, or a solicitation to buy or sell any security.